



PoA Consensus Mechanism Based on Verifiable Random Functions and BLS

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ABSTRACT

Proof of Authority (PoA) consensus algorithm mainly consists of authorized nodes to participate in each round of consensus, and the information of authorized nodes is public to a large extent, which is prone to problems such as centrality as well as predictive attacks on out-of-block nodes. To this end, this paper proposes a VB-PoA consensus algorithm based on verifiable random functions and BLS aggregated signatures. Firstly, improve the participation of nodes in each round of consensus process, select block nodes from ordinary nodes, and authorize nodes as the verification nodes of each round of consensus to avoid causing centralization. Secondly, a verifiable random function is used to randomly select the blocking nodes for blocking to avoid predictive attacks on the blocking nodes in the consensus. Finally, the aggregated signature algorithm is used to aggregate the verification information into the block header to ensure the security and correctness of the block. The experimental results show that the VB-PoA algorithm has a higher degree of node participation in the consensus process, can correctly and effectively randomly select the block nodes, and has a high degree of security.

CCS CONCEPTS

• **Security and privacy** → Cryptography; Symmetric cryptography and hash functions.

KEYWORDS

verifiable random function, aggregate signature algorithm, PoA consensus algorithm

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1 INTRODUCTION

Blockchain [1] has received attention and research from a wide range of scholars due to its decentralization, anonymity, immutability, and traceability, which does not depend on any centralized

institution. The mechanism by which each node in a distributed system agrees on the data or information present is known as the consensus of the nodes, and consensus algorithms [2] are essential for blockchain to work properly and stably.

Practical Byzantine Fault Tolerance (PBFT) is a consensus algorithm proposed by Castro et al [3], which consists of a specific set of nodes that form a consensus node set and take turns to block each other to ensure that the system works properly. The emergence of PBFT, although solves the problem of Byzantine Fault Tolerance in the traditional distributed system [4], there are still some shortcomings such as not being able to dynamically sense the joining or exiting of nodes; the master node rotates according to the numbered rotation, cannot effectively avoid malicious nodes and other problems. Bitcoin [5] was proposed by Satoshi Nakamoto in 2008, the algorithm used is Proof of Work (PoW), the algorithm system of each node through a large number of calculating the value of the nonce to obtain the power of out of the block. However, the large number of computations not only consumes many resources, but also the complete execution time of the transaction is too long. Quantum Mechanic [6] proposed Proof of Stake (PoS) consensus algorithm, the algorithm adopts voting for consensus, only to get more than half of the nodes agree to get the power of the current round of consensus block, reducing the waste of resources of the arithmetic competition, but it is easy to cause the accumulation of the coin age, and the design of a reasonable rewards and penalties mechanism is also more difficult. Authorized Proof of Stake algorithm [7] (Delegated Proof of Stake, DPoS) was proposed in 2013, which selects a specific number of nodes as representative nodes to take turns to pack the transaction requests and generate consensus blocks. Although DPoS compensates well for the shortcomings of PoW and PoS, its efficiency is improved at the expense of a certain degree of centralization.

Proof of Authority (PoA) algorithm [8] was proposed by Gavin Wood et al, mostly used in coalition chain [9], only the authorized nodes have the authority to exit blocks, sign and upload the chain, ordinary nodes want to participate in the consensus can only be through the election to become an authorized node. In the POA algorithm, the authorized nodes use a rotational approach to complete the selection of the blocking node, which results in a situation where the attacker can easily predict the blocking node in the next round of the consensus process, maliciously attacking the blocking node, which leads to the failure of the blocking. In addition, ordinary nodes cannot directly participate in the consensus process, which is prone to the problem of centralization [10].

In this paper, we propose a VB-PoA consensus algorithm based on verifiable random functions [11] and BLS aggregated signatures [12]. Firstly, through the ordinary nodes to run for the block node,

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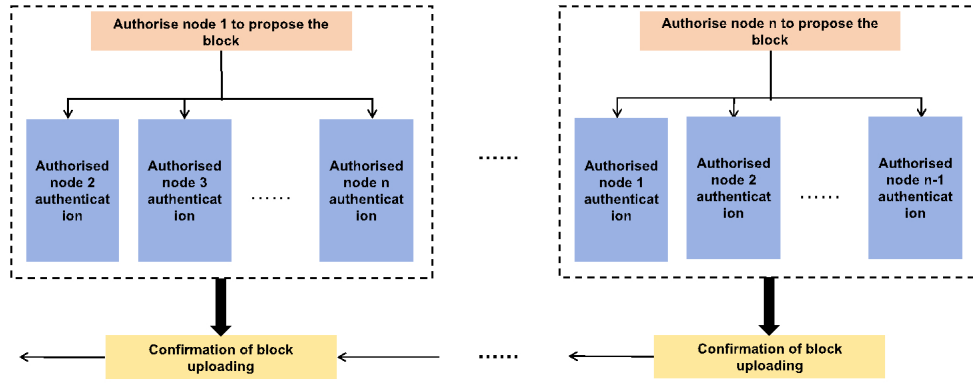


Figure 1: PoA consensus flowchart.

authorized nodes to verify the block header and block, to expand the scope of nodes participating in the consensus, to improve node participation, and to reduce the tendency of centralization. Secondly, the selection mechanism of block-out nodes in POA algorithm is improved, ordinary nodes use verifiable random function VRF to generate random numbers and proofs, and the ordinary node that generates the smallest random number becomes the block-out node of the current round of consensus, which ensures the randomness and unpredictability in the process of block-out node selection and avoids prediction attacks by malicious nodes. Finally, using the aggregated signature algorithm, the verification information is aggregated and put into the block header to ensure the correctness and security of the consensus process.

2 RELEVANT KNOWLEDGE

2.1 The PoA consensus algorithm

Proof of Authority (PoA) algorithm was proposed by Gavin Wood in 2017 and is suitable to be applied in private chains, where node roles are divided into two types: authorized nodes and ordinary nodes. Authorized nodes have the authority to exit blocks, sign and uplink; ordinary nodes do not have these authorities, but have the opportunity to transform into authorized nodes. An ordinary node can become an authorized node by getting more than 50% of the election votes and thus has the authority to participate in the consensus. The PoA consensus process is shown in Figure 1. PoA consensus algorithm usually includes steps such as node authentication, node consensus, and node determination.

2.2 Verifiable Random Functions

Verifiable Random Function (VRF) is a cryptographic function proposed by Silvio Micali to generate unpredictable random numbers that can be verified at the same time. VRF generates random numbers and proofs based on data inputs, and other people can verify the generated results. The input consists of two parts, one is the user's own private key and the other is the publicly available seed S . The output is the random numbers and the corresponding proofs; other people who have the public key can verify that the random numbers and proofs are correct, but cannot infer how the data was generated. VRF is random, verifiable and unique [13].

VRF is overall divided into two parts: generating proofs and verifying proofs [14], the generating proof process uses the private key SK and the seed S as inputs to the VRF function:

$$P = VRF_proof(SK, S) \quad (1)$$

$$R = VRF_hash(P) \quad (2)$$

where SK is the private key, S is the random number, P is the generated proof, and R is the generated random number. The verification process utilizes the public key PK , the seed S , and the proof P to test whether the proof P is based on the proof generated by S and whether R is the correct output, and if so returns legal, otherwise illegal.

$$VRF_verify(PK, S, P) \quad (3)$$

2.3 Aggregate Signature Algorithm

In 2004, Dan Boneh proposed a signature scheme named BLS aggregated signature algorithm [15]. The main process consists of four parts: initial process, key generation, aggregated signature and verification of signature.

Initial procedure: G_1, G_2 is a multiplicative cyclic group of order p , the generators are g_1, g_2 , respectively, e is a bilinear mapping [16]: $G_1 * G_2 \rightarrow G_T$, a secure hash function $H : M \rightarrow G_1$, and the public arguments are $(G_1, G_2, G_T, p, g_1, g_2, e, H)$.

Key Generation: choose a random number α and set $h \leftarrow g_1$, $\alpha \in G_1$ to compute the private key $SK = \alpha$ and public key $PK = h$.

Aggregate signature: curve hash [17] on message m to get $H(m)$, signing using private key; for a given tuple (PK_i, m, σ_i) , where $i = 1, \dots, n$, $\sigma_1, \dots, \sigma_n \in G_1$ is aggregated into short aggregate signatures by Equation 4).

$$\sigma_i = SK_i \times H(m) \quad (4)$$

Verify Signature: when verifying the aggregated signature $\sigma \leftarrow \sigma_1, \dots, \sigma_n \in G_1$, use the public key PK to verify, and the output result of TRUE indicates that the verification has been passed, otherwise, it means that the verification has not been passed.

$$e(g_1, \sigma) = e(PK_1, \dots, PK_n, H(m)) \quad (5)$$

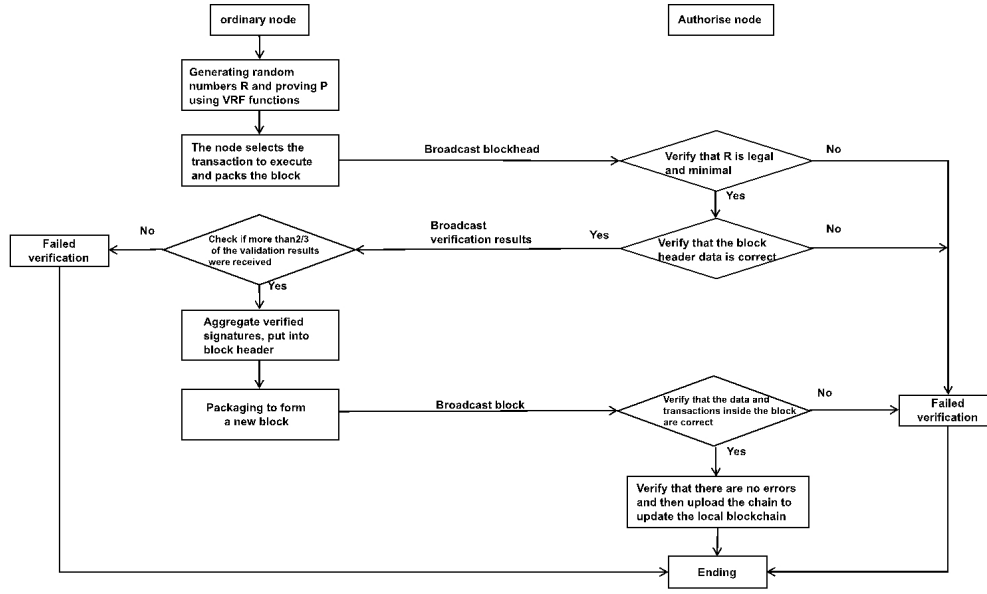


Figure 2: Flowchart of VB-PoA consensus algorithm.

3 ALGORITHM DESIGN

The VB-PoA consensus algorithm based on verifiable random function and aggregated signature proposed in this paper can be divided into the following processes. The flowchart of VB-PoA algorithm is shown in Figure 2.

Determine the nodes involved in the consensus: only authorized nodes participate in each round of consensus in PoA algorithm, the improved VB-PoA consensus algorithm increases the participation of ordinary nodes in the consensus, which is not limited to authorized nodes, and expands the participation of nodes.

Randomly select out block nodes: PoA consensus process uses a rotating block out of the authorization node set (ValidatorSet) to select the block out of each round of consensus node, but with the increase in the number of consensus, the authorization node's information has been made public to a progressively greater extent, and this public information will lead to the consensus process of the block out of the node easily predicted attacks. To address this problem, the improved VB-PoA consensus algorithm adopts the verifiable random function VRF to randomly select the nodes for blocking, which not only ensures that the consensus nodes in each round of the blocking process cannot be predicted to be attacked by malicious nodes, but also realizes random selection as well as can be verified. The process of randomly selecting nodes for block out is shown in Algorithm 1

Ordinary nodes, as potential block-out nodes, firstly use their own private key SK and a network-wide known parameter seed as inputs to the VRF function, which is computed to generate a random number R and a proof P. Next, they select transactions from the transaction pool for execution and package them to generate a candidate block, in which they will put the VRF generated random number R and the proof P into the block header of the candidate block, which will then be broadcasted to multiple authorized nodes.

Algorithm 1 Random selected block node algorithm.

Input: node private key SK, hash value seen of the previous block

Output: random number R, proof P

1. $x \leftarrow sk$, $H = \text{hash}_{\text{to_curve}}(x * G)$
2. $Gm = x * H$
3. $R = \text{nonce_generation}(sk, (\text{point_to_string}(H)))$
4. $c = \text{hash}_{\text{points}}(H, Gm, R * G, R * H)$
5. $s = (R + c * x) \bmod q$
6. $p = \text{point_to_string}(Gm) || \text{int_to_string}(c, n) || \text{int_to_string}(s, q)$
7. Return to R, P

Verify and confirm the outgoing block node: when an authorized node receives a candidate block, it first verifies that the identity of the potential outgoing node is correct, followed by verifying the data within the block header of the candidate block.

The authorized node verifies that the random number R and the proof P are correctly generated by the VRF function and that the data has not been tampered with through the public key PK of the potential block-out node. If the validation fails, the corresponding block header is directly discarded; if the validation passes, the smallest random number R is next selected by comparison, and the potential outgoing node that generates the smallest random number R will become the outgoing node of the current consensus round.

The authorized node verifies the data inside the block header of the candidate block, if the verification fails, the block header is discarded directly; if the verification passes, the voting signature is performed and the voting result is broadcasted. The process of verifying the outgoing block node is as in Algorithm 2

Ordinary nodes check the number of verification messages received. If it receives less than 1/3, it indicates that it is not an

Table 1: Experimental environment configuration.

Version information	Version number
Operating System (OS)	Win10
CPU	Intel(R) Core(TM)i5-9400
Internal storage	8G
Go	GoLand 2022.2.4

outgoing node for this round of consensus and does not do anything. If more than $1/3$ is less than $2/3$, then other nodes must also not receive more than $2/3$ of the verification messages, indicating that there is an abnormality in this round of consensus, then broadcast a request message (NEW- ROUND, V, R, P) requesting to open a new round of block-out node election. Where V is the collection of verification messages received by the outgoing node, R and P are random numbers and proofs generated by VRF. After all nodes receive the request message, they verify the signature and determine whether the message in V is legitimate or not, and open a new round of consensus after the verification passes. If it receives more than $2/3$ of the verified nodes' voting signatures, it successfully becomes the outgoing node for this round of consensus.

Algorithm 2 Verify the block node.

Input: Node public key PK, P, hash of last block seed

Output: TRUE/FALSE

1. $D = \text{decode}_{\text{proof}}(P)$
2. Deconstructing D content: $(Gm, c, s) = D$
3. Verify D legitimacy, if not stop the operation
4. $H = \text{hash}_{\text{to curve}}(pk, \text{seed})$
5. $U = s * G - c * pk$
6. $V = s * H - c * Gm$
7. $c' = \text{hash}_{\text{points}}(H, Gm, U, V)$
8. Verify that $c == c'$ is legal, return true if it is legal, return false if it is not.

Consensus on the chain: the outgoing node first uses the aggregated signature BLS algorithm to aggregate the voting signatures of the authorized nodes into a new signature that is added to the block header; then the new block containing the aggregated signature is broadcast to other nodes in the network. After receiving the block, the nodes in the network verify all the transactions and data in the block, and if all the transactions are legal, they add them to the local blockchain and update the local blockchain; if one of the transactions is not legal, they reject the block.

4 EXPERIMENTAL ANALYSIS

In this paper, the VB-PoA algorithm is implemented by Go language, and the consensus simulation experiment is carried out by using the blockchain test chain constructed by multiple nodes, and the experimental environment is configured in Table 1.

Table 2: Average running time of the first 200 VRF algorithms.

Algorithm	Version number
Random selected block node	0.912
Verify the block node	0.998

Table 3: Results of the first 10 block-out node selections.

Rounds	Block node
1	D
2	J
3	A
4	E
5	B
6	J
7	C
8	D
9	H
10	F

4.1 Verifiable stochastic function analysis

In this experiment, 20 nodes are set up, among which there are 10 ordinary nodes participating in the election of the outgoing nodes, which are A~J; and there are 10 authorized nodes carrying out the verification, which are 0~9. The VRF randomly selected block out node is tested 200 times and the time of randomly selected block out node and verified block out node is recorded in each run, and then the average running time of the algorithm of randomly selected block out node and verified block out node is calculated as in Table 2. From the table, it can be seen that the VRF function selects and verifies the blocking nodes in milliseconds on average each time, so the improved blocking node selection and verification does not have a significant impact on the time of the whole consensus process, and will not affect the whole blocking process.

Record the block-out nodes selected by VRF for the first 10 times, as shown in Table 3. it can be seen that the block-out nodes selected by using the VRF function in each round of the consensus process do not have any relationship with the block-out nodes selected by the previous round of the sub-consensus, and the whole process has a strong randomness and unpredictability. Moreover, with the increase of the number of consensus, the information of the block-out nodes will not be leaked.

4.2 Security Analysis

Firstly, VB-PoA uses the VRF function, whose randomness and unpredictability can ensure that in the process of selecting and verifying out-of-block nodes, the information of nodes is not leaked, so that the malicious nodes can not be predicted and attacked, and the security of the node selection process is guaranteed to the

maximum extent. Firstly, VB-PoA uses the VRF function, whose randomness and unpredictability can ensure that the information of nodes is not leaked in the process of selecting and verifying the block-out nodes, so that the malicious nodes can not be predicted and attacked, and the security of the node selection process is guaranteed to the maximum extent. Secondly, authorized nodes and ordinary nodes have the authority to participate in consensus only through identity verification, which raises the threshold of node joining. Finally, in the verification node verification stage, the verification signatures of all authorized nodes are aggregated using the aggregated signature algorithm and added to the block header to record the behavior of each outgoing node, which facilitates the supervision of other nodes and ensures that the outgoing nodes do not behave maliciously in the consensus process.

5 CONCLUDING REMARKS

Consensus algorithm is an important mechanism to ensure the normal operation of blockchain, this paper firstly introduces blockchain related knowledge, and then proposes VB-PoA consensus algorithm based on Verifiable Random Function (VRF) and Aggregate Signature (BLS) for the problems existing in PoA consensus algorithm, randomly selects block nodes through Verifiable Random Function (VRF), to ensure randomness and unpredictability in the process of node selection. The VRF randomly selects block nodes to ensure the randomness and unpredictability in the selection process of block nodes and prevent them from being predicted and attacked by malicious nodes; it expands the degree of node participation in the consensus and introduces ordinary nodes to participate in the election of block nodes, so as to prevent centralization; and it supervises the behavior of block nodes through the aggregation signature algorithm by aggregating the validation information into block header, so as to ensure the security. After experimental verification, the improvement of this paper did not have a great impact on the original block generation time, can randomly select the block nodes, and according to a certain time interval normal block; and through the analysis of the security of this algorithm, this scheme than the PoA greatly improves the consensus process out of the block nodes of the unpredictability and randomness, reduces the

nodes of the leakage of information, and has a high degree of security. At present, this optimization is only an improvement in the security enhancement of PoA consensus algorithm, and in the future, it can conduct in-depth research on its transaction performance, continuously improve the consensus algorithm and enhance its efficiency.

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